ANALYSIS II — SHEET 1 (for Week 2 tutorials)
Limits and Continuity on Intervals

1. Suppose that $I \subseteq \mathbb{R}$ is non-empty, bounded and has the *interval property*: that is, if $x, y \in I$ then $[x, y] \subseteq I$. Show carefully that there exist $a, b \in \mathbb{R}$ such that I is one of

$$(a, b)\,, \qquad (a, b], \qquad [a, b), \qquad [a, b]\,.$$

2. (i) Prove, from the definition, that if $f\colon \mathbb{R} \to \mathbb{R}$ is continuous at x_0, then $|f|$ is continuous at x_0.
(ii) Show that the converse of (i) is untrue.
(iii) Prove that if f and g are continuous at x_0, then $\max\{f, g\}$ and $\min\{f.g\}$ are continuous at x_0.

3. Prove carefully from definitions that if

$$\lim_{x\to x_0} f(x) = y_0, \qquad \lim_{y\to y_0} g(y) = L, \qquad g(y_0) = L,$$

then $g(f(x)) \to L$ as $x \to x_0$. Give an example to show that the condition $g(y_0) = L$ is necessary for this conclusion.

4. (i) Find a function $f\colon \mathbb{R} \to \mathbb{R}$ which is discontinuous at all points of the set $\left\{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$ but is continuous everywhere else.
(ii) Find a function $f\colon \mathbb{R} \to \mathbb{R}$ which is discontinuous at all points of the set $\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$ but is continuous everywhere else.

5. Determine if the following limits exist and evaluate those that do.

$$\lim_{x\to\infty} \frac{x^2}{x^3+1}, \qquad \lim_{x\to\infty} \frac{x \sin x}{x+1}, \qquad \lim_{x\to\infty} \frac{\log x}{x},$$
$$\lim_{x\to 0} \frac{\sin(x^2)}{\sin^2 x}, \qquad \lim_{x\to 1} \frac{\sin(x-1)}{x^2-1}, \qquad \lim_{x\to 0} \sin(x) \sin\left(\frac{1}{x}\right).$$

6. The function $f\colon \mathbb{R} \to \mathbb{R}$ is *additive* meaning

$$f(x+y) = f(x) + f(y) \qquad \text{for all } x, y \in \mathbb{R}.$$

(i) Show that $f(nx) = nf(x)$ for all $n \in \mathbb{Z}$ and $x \in \mathbb{R}$.
(ii) Deduce that $f(qx) = qf(x)$ for all $q \in \mathbb{Q}$ and $x \in \mathbb{R}$.
(iii) Show that if f is continuous, then there is $c \in \mathbb{R}$ such that $f(x) = cx$ for all $x \in \mathbb{R}$.
(iv) Show that if f is continuous at 0, then f is in fact continuous at all $x \in \mathbb{R}$.
(v) (Optional) (ii) shows that f is additive if and only if f is a linear map of $\mathbb{R}$ as a vector space over $\mathbb{Q}$. What is $\dim_{\mathbb{Q}} \mathbb{R}$? Assuming a basis

$$\mathcal{B} = \{r_i \mid i \in I\}$$

for $\mathbb{R}$ over $\mathbb{Q}$ exists, describe a *discontinuous* additive function $f\colon \mathbb{R} \to \mathbb{R}$.

Extension Exercises (Optional)

7. Let $\lfloor x \rfloor$ denote the *floor* or *integer part* of x. Determine the points of continuity of the following functions $f\colon \mathbb{R} \to \mathbb{R}$.

(i) $f(x) = \lfloor x \rfloor$.
(ii) $f(x) = x\lfloor x \rfloor$.
(iii) $f(x) = \lfloor \sin x \rfloor$.
(iv) $f(x) = \lfloor 1/x \rfloor$ when $x \neq 0$ and $f(0) = 0$.
(v) $f(x) = x\lfloor 1/x \rfloor$ when $x \neq 0$ and $f(0) = 1$.

8. Determine if the following limits exist and evaluate those that do.

$$\lim_{x\to 0} \frac{\sqrt{1+x} - \sqrt{1+3x}}{x}, \qquad \lim_{x\to 1} \frac{\sqrt{x} - 1}{x - 1}, \qquad \lim_{x\to 0^-} \frac{1}{1 - e^{1/x}}. \qquad \lim_{x\to 0^+} \frac{1}{1 - e^{1/x}}, \qquad \lim_{x\to 0} \frac{1}{1 - e^{1/x}}.$$

9. (i) Let $f\colon \mathbb{R} \to \mathbb{R}$ be a continuous function with periods $T_1, T_2 > 0$ such that T_1/T_2 is irrational. Show that f is constant.

(ii) Let $f\colon \mathbb{R} \to \mathbb{R}$ be non-constant, continuous and periodic. Show that f has a smallest positive period.

10 *Thomae's function* $f\colon [0,1] \to \mathbb{R}$ (aka the popcorn function) is defined by

$$f(x) = \begin{cases} \frac{1}{q} & \text{if } x = \frac{p}{q} \in \mathbb{Q} \text{ in lowest terms;} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

For what values of x is $f(x)$ continuous? Justify your answer.

11. (i) Find a continuous function $f\colon (0,\infty) \to (0,\infty)$, other than the identity function, such that $f(2x) = 2f(x)$ for all x and $f(1) = 1$.

(ii) Find a continuous function $g\colon (0,1) \to (0,1)$, other than the identity function, such that $g(x^2) = g(x)^2$ for all $x \in (0,1)$.

12. Let $f\colon \mathbb{R} \to \mathbb{R}$ be such that for every real p the limit

$$g(p) = \lim_{x\to p} f(x)$$

exists and is finite. Show that $g(x)$ is continuous. [Hint: given $p_n \to p$, show that $g(p_n) \to g(p)$ by approximating $g(p_n)$ by suitable $f(q_n)$.]

Does f have to be continuous?